Mathematical Modelling at Times of Pandemic: The Good, the Bad, and the Ugly

Andrea Saltelli

Seminar
Models of pandemics: What should we do differently next time?

UPF-Barcelona School of Management
February 03, 2026

Journal of Evaluation in Clinical Practice
International Journal of Public Health Policy and Health Services Research

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COMMENTARY | Open Access

What did COVID-19 really teach us about science, evidence and society?

Andrea Saltelli, Joachim P. Sturmberg, Daniel Sarewitz, John P. A. Ioannidis

First published: 06 June 2023 | https://doi.org/10.1111/jep.13876

"COVID-19 policies allocated sacrifice, privation and suffering across all walks of society [but] radically different responses from nation to nation—from draconian lockdowns, to relatively permissive and flexible pandemic regimes—made obvious to all that the value of scientific evidence was to support what was politically desirable and possible in different contexts

Mostly provided by models

COMMENT · 24 JUNE 2020

Five ways to ensure that models serve society: a manifesto

Pandemic politics highlight how predictions need to be transparent and humble to invite insight, not blame.

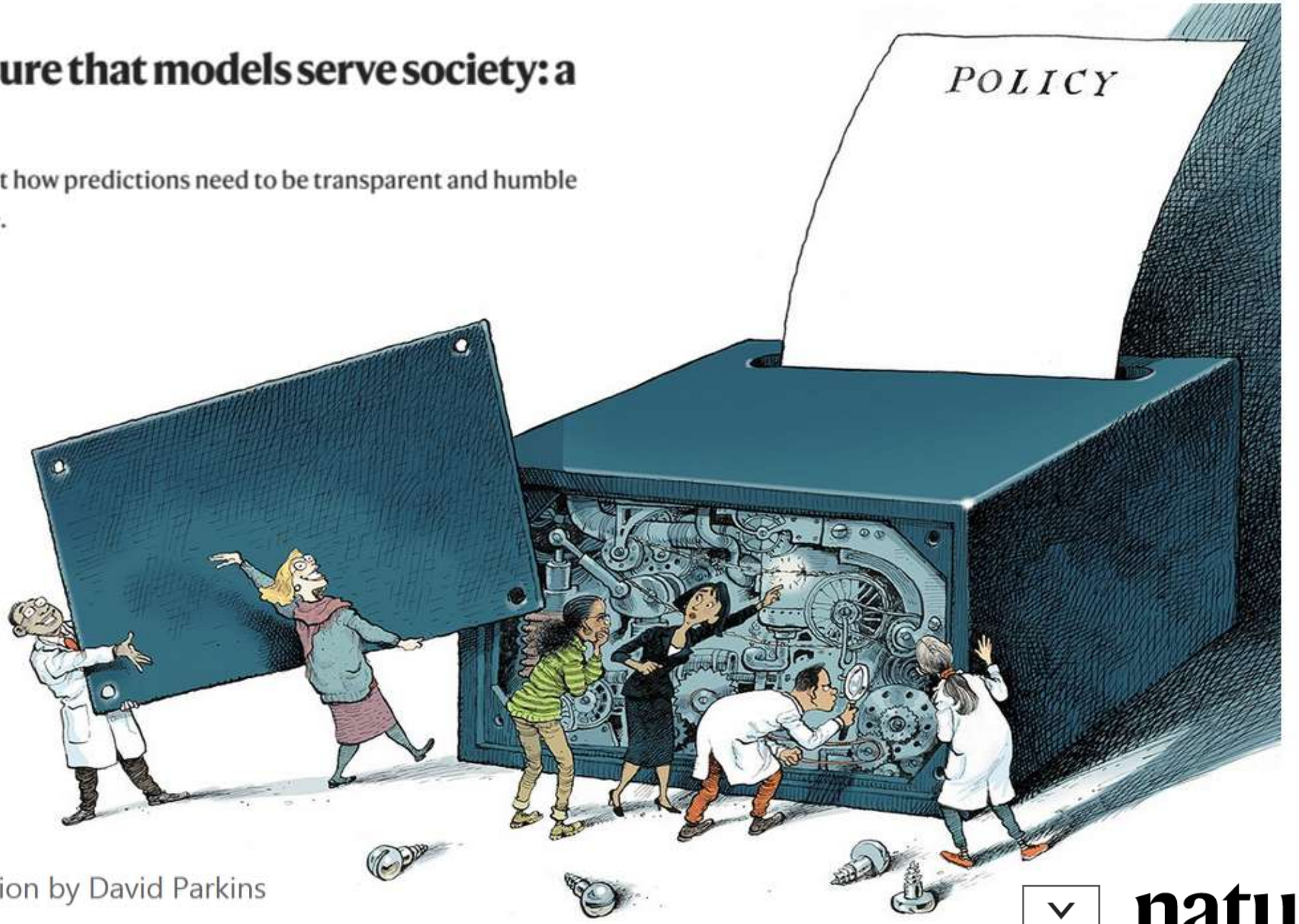
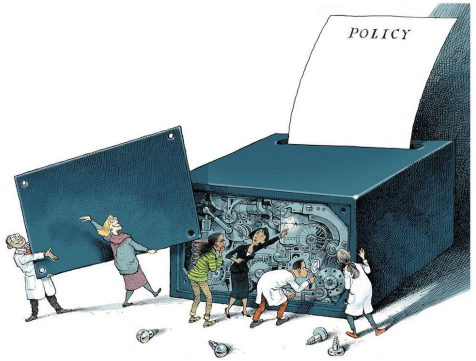


Illustration by David Parkins

nature

Models become visible and powerful



The New York Times

Behind the Virus Report That Jarred the U.S. and the U.K. to Action

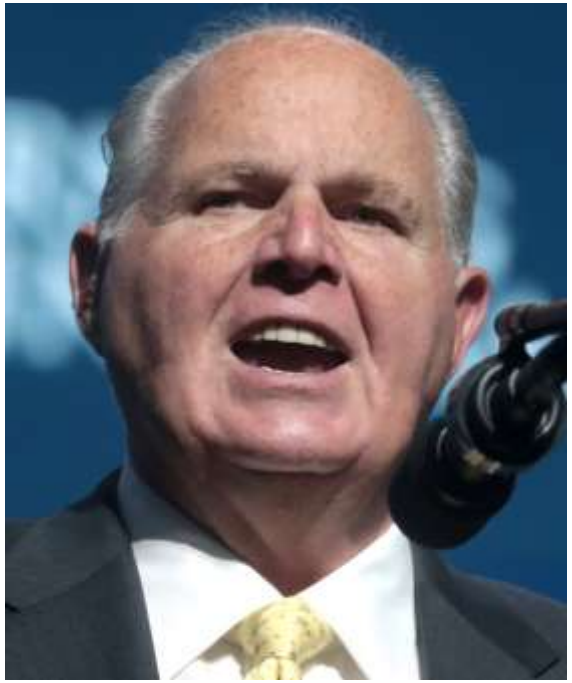
It wasn't so much the numbers themselves, frightening though they were, as who reported them: Imperial College London.

Landler, Mark, and Stephen Castle. 2020. Behind the Virus Report That Jarred the U.S. and the U.K. to Action - The New York Times.

Models become controversial, when questions of urgency, stakes, values and uncertainty collide

"Wild-Ass Covid numbers … The minute I hear anybody start talking about models and modeling, I blanch"

Rush Limbaugh



Rhodes, Tim, and Kari Lancaster. 2020. "**Mathematical Models as Public Troubles in COVID-19 Infection Control: Following the Numbers**", Health Sociology Review 1-18. doi: 10.1080/14461242.2020.1764376

Mind the assumptions

Assess uncertainty and sensitivity

Mind the hubris

Complexity can be the enemy of relevance

Mind the framing

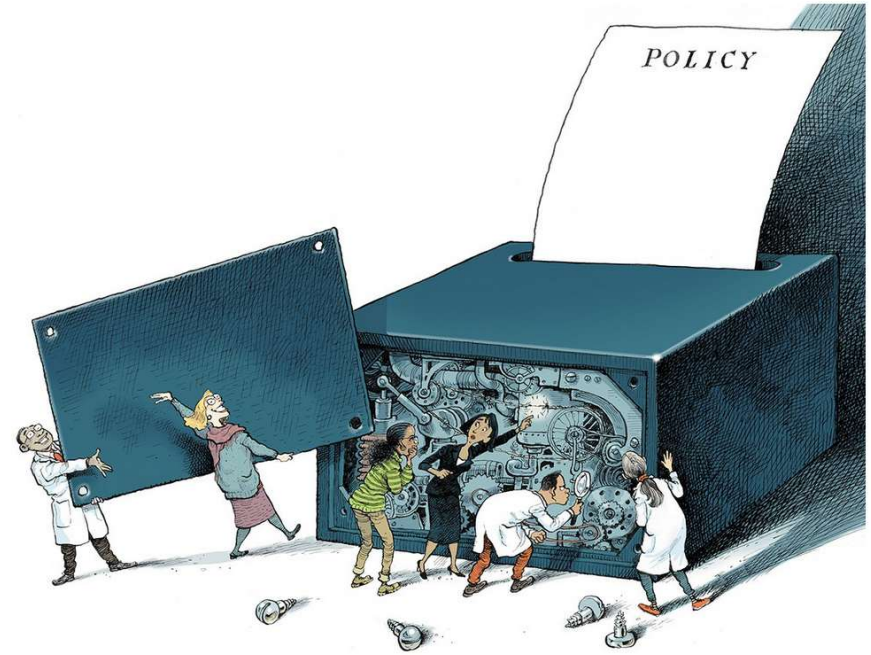
Match purpose and context

Mind the consequences

Quantification can backfire.

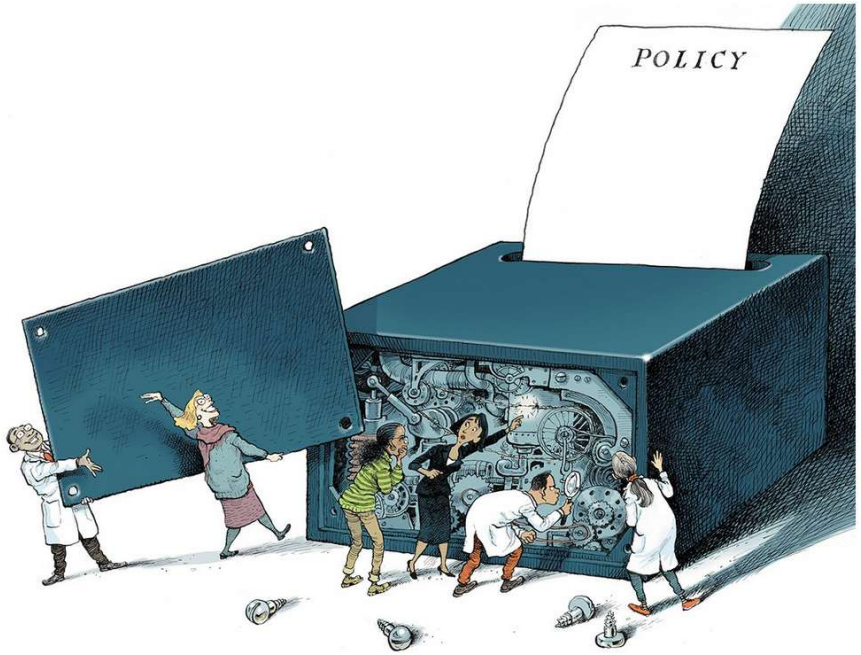
Mind the unknowns

Acknowledge ignorance



Mind the assumptions

Assess uncertainty and sensitivity



… models require input values for which there is no reliable information…

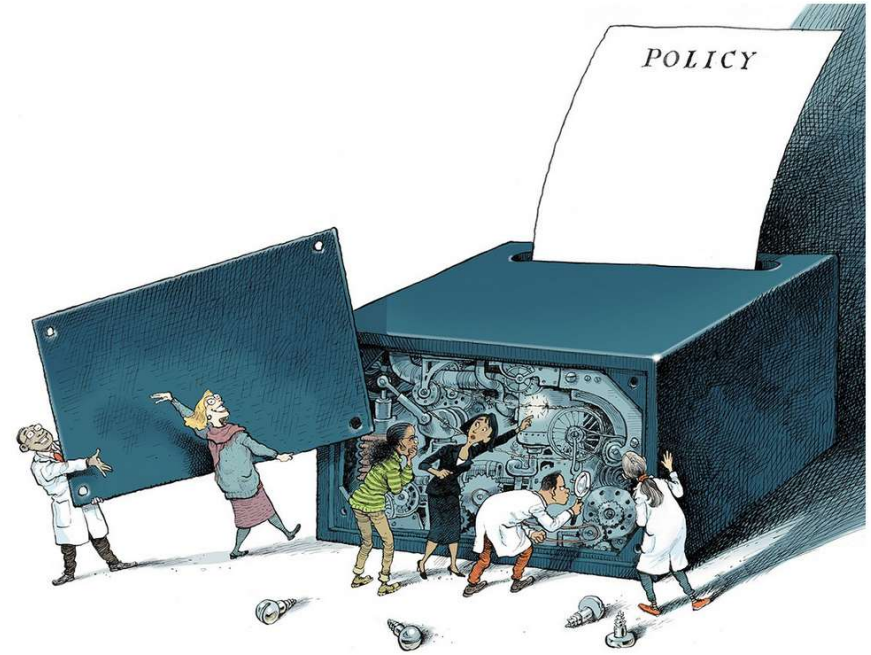
…global uncertainty and sensitivity analyses are often not done. Anyone turning to a model for insight should demand them …

Mind the assumptions

Assess uncertainty and sensitivity

Mind the hubris

Complexity can be the enemy of relevance



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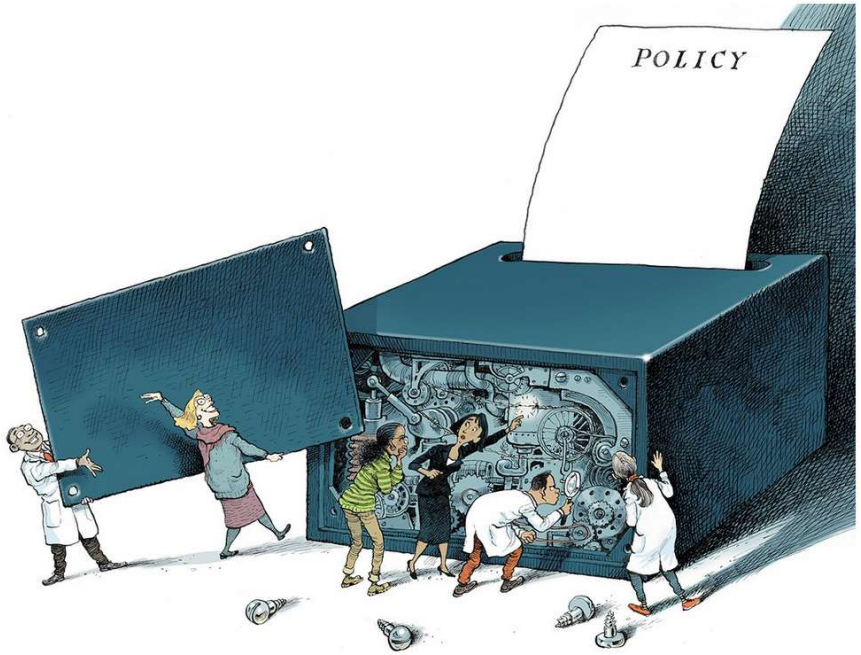

… models will reflect the interests, disciplinary orientations and biases of the developers…

SUPPLEMENTARY INFORMATION

1. Additional information and references >260 references

Watch philosopher Eric Winsberg on YouTube, February 2022

Who takes the viewpoint of blue-collar workers?

https://www.youtube.com/watch?v=_cgCTK17ics

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My recent work on models: a state of exception?

… descending from a mix of opacity and epistemic authority

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Research Article

Mathematical models: a state of exception

Andrea Saltelli, Arnald Puy & Monica Di Fiore

Pages 862-879 | Received 09 Jan 2024, Accepted 07 May 2024, Published online: 11 Jun 2024

Cite this article https://doi.org/10.1080/02692171.2024.2365727 Check for updates

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Environmental Modelling & Software

Volume 188, April 2025, 106430

Models and the common good ☆

Andrea Saltelli ✉

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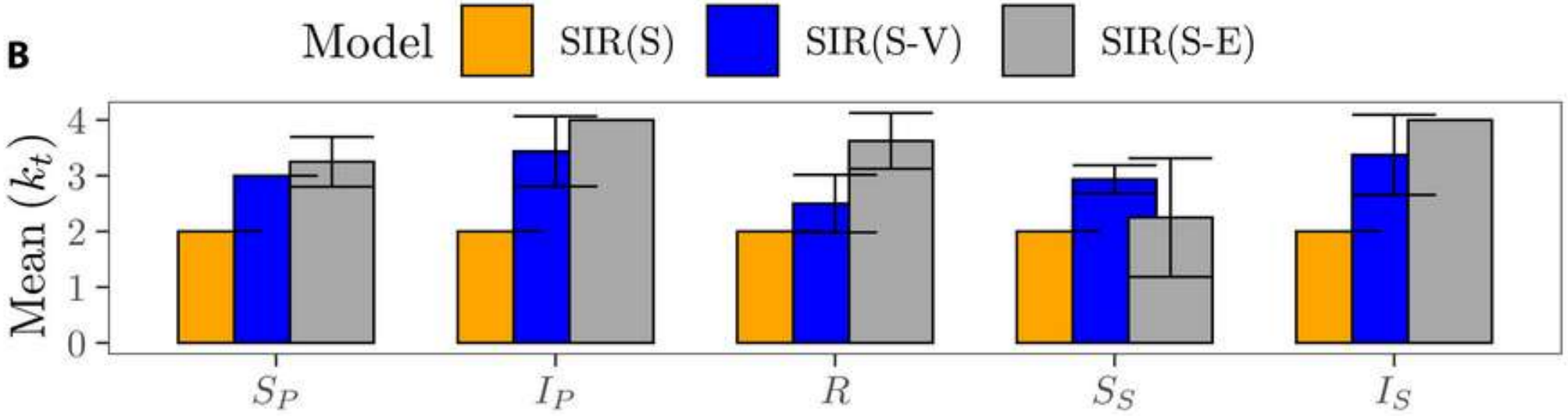
Science Advances

RESEARCH ARTICLE | MATHEMATICS

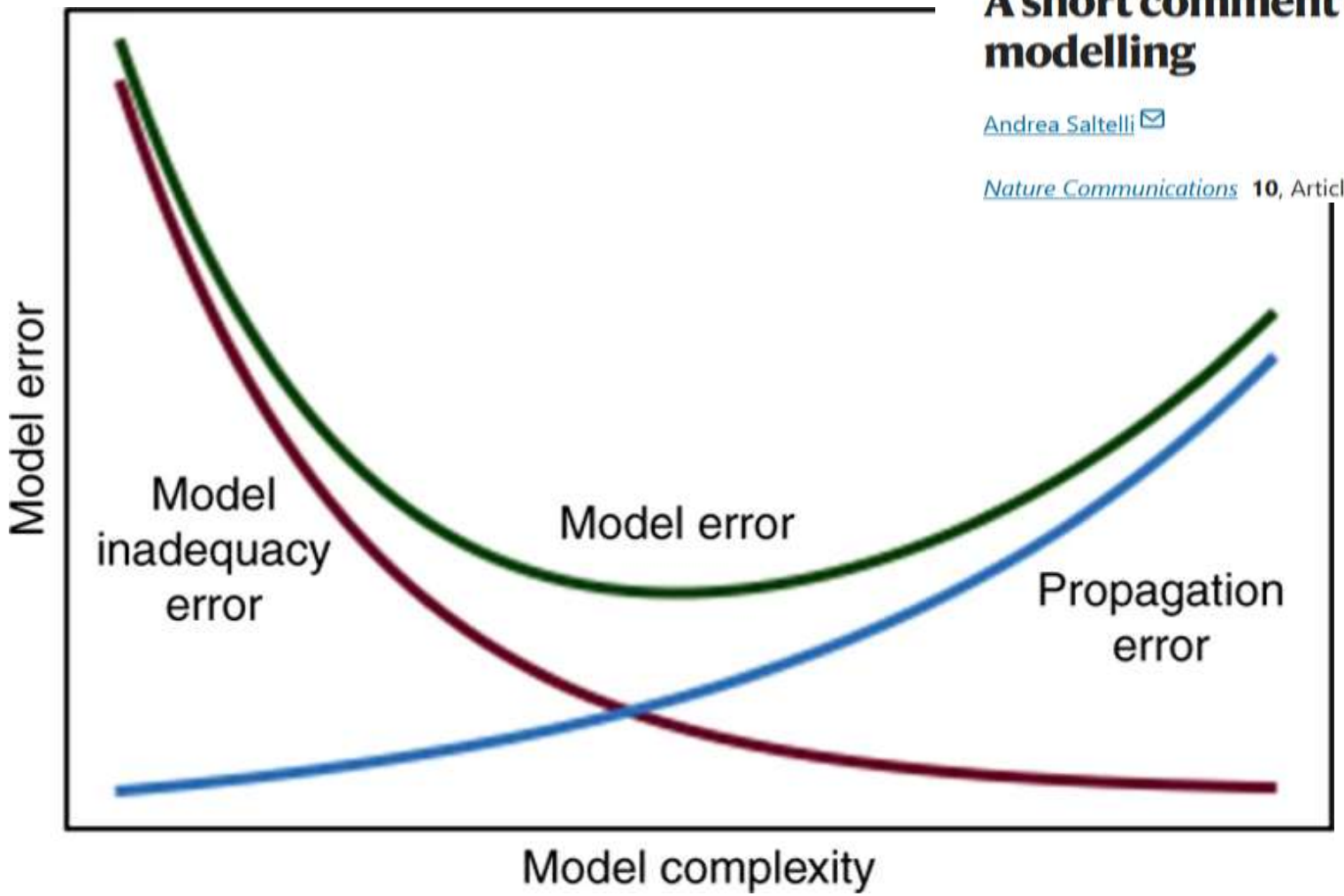
Models with higher effective dimensions tend to produce more uncertain estimates

ARNALD PUY, PIERFRANCESCO BENEVENTANO, SIMON A. LEVIN, SAMUELE LO PIANO, TOMMASO PORTALURI, AND ANDREA SALTELLI Authors Info & Affiliations

Studying SIR models of increasing complexity



O'Neil conjecture at work



nature communications

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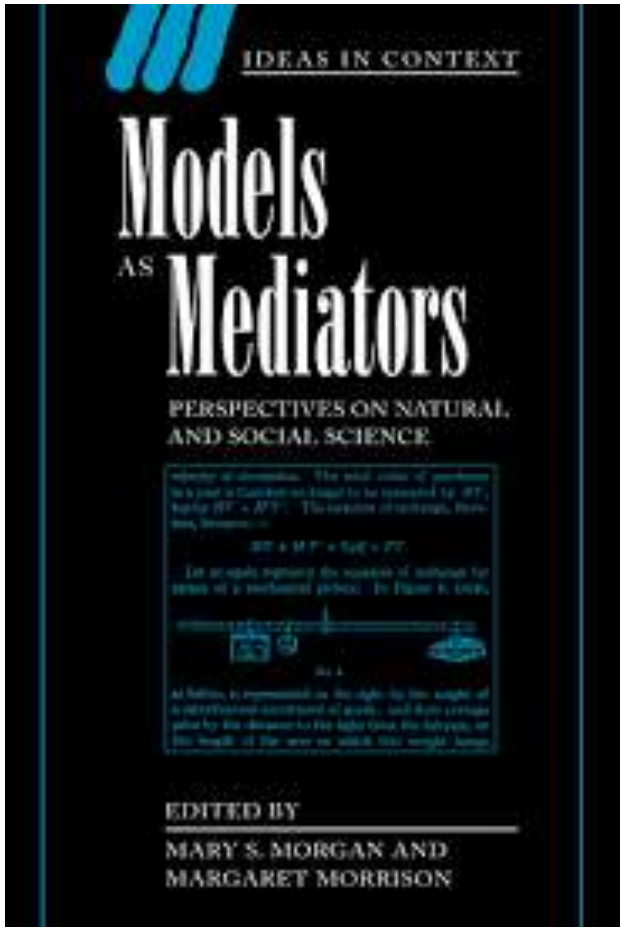
A short comment on statistical versus mathematical modelling

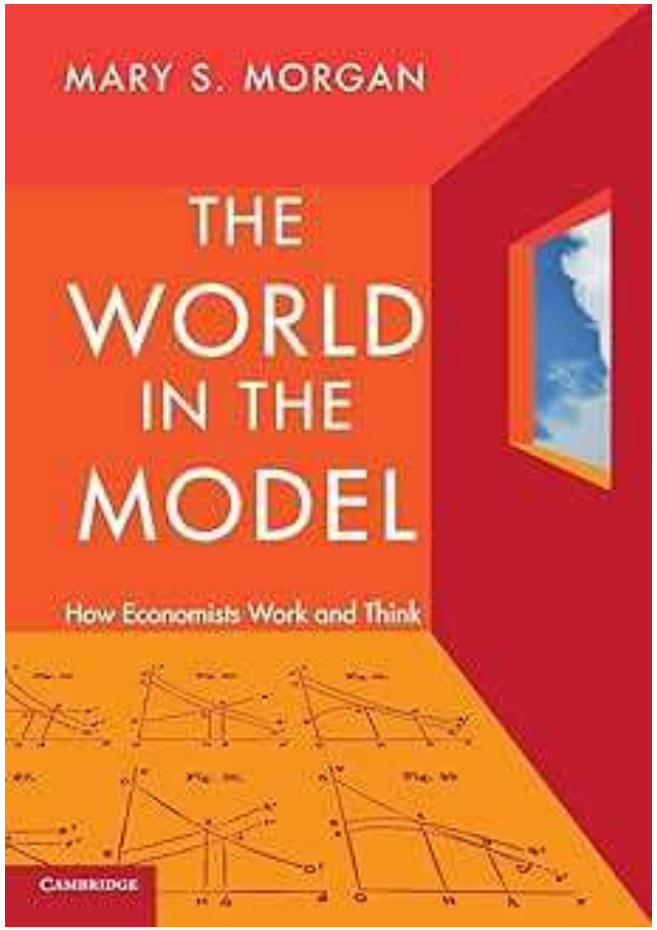
Andrea Saltelli

Nature Communications **10**, Article number: 3870 (2019) | Cite this article

Rare incursions from the social sciences
(when compared to other instances of quantification)

➔ Who defends society from models?





Consequences descending from state of exception

Models have a strong grip in policy

Models have their own political economy - economicism, solutionism, reductionism, transforming of the qualitative into quantitative (Stirling, 2023a, 2023b).

The percentage of non-reproducible studies in the field of clinical medical research could reach 85% (Chalmers and Glasziou, 2009). Nobody can provide a similar figure for mathematical modelling.

'Navigating the political' (van Beek *et al.* 2022)

Acting as chameleons, jumping across contexts, Pfleiderer (2020).

Source: National Geographic

Chalmers, Iain, and Paul Glasziou. 2009. 'Avoidable Waste in the Production and Reporting of Research Evidence'. The Lancet 374 (9683): 86–89.

Pfleiderer, Paul. 2020. 'Chameleons: The Misuse of Theoretical Models in Finance and Economics'. Economica 87 (345): 81–107.

Stirling, Andy. 2023. 'Against Misleading Technocratic Precision in Research Evaluation and Wider Policy – A Response to Franzoni and Stephan (2023), "Uncertainty and Risk-Taking in Science"'. Research Policy 52 (3): 104709.

van Beek, Lisette, Jeroen Oomen, Maarten Hajer, Peter Pelzer, and Detlef van Vuuren. 2022. 'Navigating the Political: An Analysis of Political Calibration of Integrated Assessment Modelling in Light of the 1.5 °C Goal'. Environmental Science & Policy 133 (July):193–202.

"At the very least, a 'red team' with similar heft to SAGE would seem appropriate as soon as a massively costly experimental intervention becomes a serious possibility. Such a team might, for example, have questioned the fairness of allowing the cost per life year saved from Covid to be many times the usual NICE threshold for approval of an intervention (about £30,000 per QALY)"

Did we do it well?

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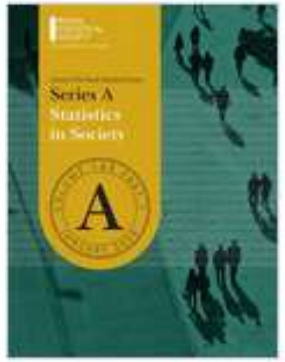

Volume 189, Issue 1
January 2026

JOURNAL ARTICLE

Some statistical aspects of the Covid-19 response

Simon N Wood, Ernst C Wit, Paul M McKeigue, Danshu Hu, Beth Flood, Lauren Corcoran, Thea Abou Jawad Author Notes

Journal of the Royal Statistical Society Series A: Statistics in Society, Volume 189, Issue 1, January 2026, Pages 1–31, https://doi.org/10.1093/jrsssa/qnaf092
Published: 01 July 2025 **Article history** ▾

BBC

Five years on: The countries that never locked down for Covid-19

Chris Baraniuk

DISPATCH

SWEDEN'S PANDEMIC EXPERIMENT

When the coronavirus arrived, the country decided not to implement lockdowns or recommend masks. How has it fared?

By Mallory Pickett

April 6, 2021

THE NEW YORKER 100

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Which country did best?

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US politics

This article is more than 9 months old

'A case study in groupthink': were liberals wrong about the pandemic?

A train conductor walks through a virtually empty Grand Central Terminal, 22 March 2020, New York. Photograph: Gary Hershorn/Getty Images

US political scientists' book argues aggressive Covid policies such as mask mandates were in some cases misguided

How about the origin of the virus?

CIA says COVID-19 more likely to have come from lab than nature — Reuters

By Erin Banco

January 27, 2025 5:31 PM GMT+1 · Updated January 27, 2025

nature

Article | Published: 26 March 2020

Identifying SARS-CoV-2-related coronaviruses in Malayan pangolins

Tommy Tsan-Yuk Lam, Na Jia, Ya-Wei Zhang, Marcus Ho-Hin Shum, Jia-Fu Jiang, Hua-Chen Zhu, … Tong, Yong-Xia Shi, Xue-Bing Ni, Yun-Shi Liao, Wen-Juan Li, Bao-Gui Jiang, Wei Wei, Ting-Ting Yu, … Zheng, Xiao-Ming Cui, Jie Li, Guang-Qian Pei, Xin Qiang, William Yiu-Man Cheung, Lian-Feng Li, … Sun, Si Qin, Ji-Cheng Huang, … Wu-Chun Cao + Show authors

Nature **583**, 282–285 (2020) | Cite this article

nature

NEWS | 20 September 2024

COVID pandemic started in Wuhan market animals after all, suggests latest study

The finding comes from a reanalysis of genomic data.

By Smriti Mallapaty

nature

NEWS | 04 December 2024

Sick animals suggest COVID pandemic started in Wuhan market

A preliminary analysis takes a closer look at genomic data collected at the market.

By Smriti Mallapaty

In COVID (as elsewhere), models influenced narrative and polices …

Leading to a dramatic collapse of trust in science/expertise

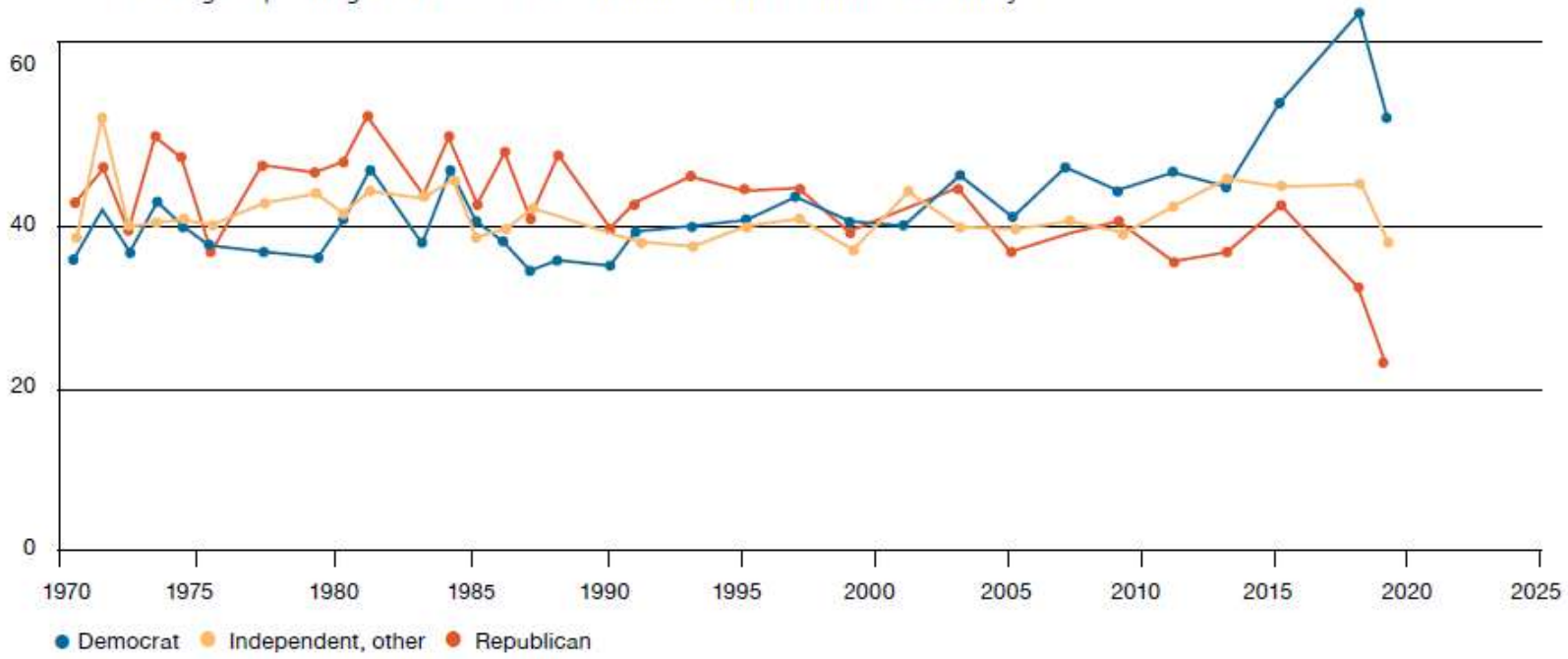
ISSUES IN SCIENCE AND TECHNOLOGY

The Strange New Politics of Science

Anthony Mills and Price St Clair. The strange new politics of science, April 2025. URL: https://issues.org/new-politics-science-mills-st-clair/.

Figure 1. **CONFIDENCE IN SCIENCE BY POLITICAL LEANING**

Percentage expressing "A Great Deal" of confidence in the scientific community.



Source: General Social Survey: Confidence in the Scientific Community. Courtesy American Enterprise Institute.

The polarization of trust in science is a complex phenomenon shaped by—and increasingly also shaping—American political identities.

ISSUES

In COVID (as elsewhere), models influenced narratives and polices …

…fostering societal polarization, as models feed into an economy of virtue signalling, where progressive elites manifest their conspicuous cognition, enraging the ‘deplorables’

https://www.youtube.com/watch?v=PCHJVE9trSM

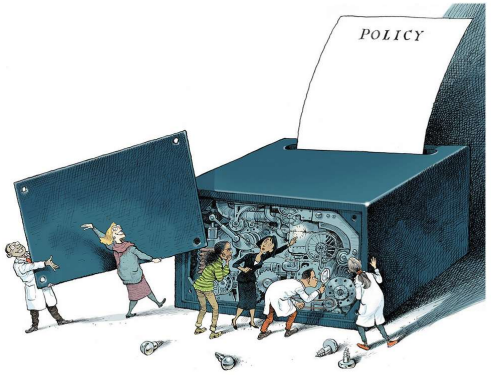
Polarization generates distrust in science, or are attitudes toward science that generate polarization?

The polarization of trust in science is a complex phenomenon shaped by—and increasingly also shaping—American political identities.

ISSUES

Be wary of ‘following the science’

Use models with parsimony and circumspection



Communicating models’ conditionality

Andy Stirling

Make room for counter-modelling (red team)

Ehsan Nabavi

www.water-alternatives.org

Volume 18 | Issue 2

Nabavi, E. 2025. Modelling as intervention technology: Science, politics, and water conflicts. Water Alternatives 18(2): 330-354

Water Alternatives

Modelling as Intervention Technology: Science, Politics, and Water Conflicts

Ehsan Nabavi

Responsible Innovation Lab, The Australian National Centre for the Public Awareness of Science, The Australian National University; and Käte Hamburger Kolleg, Cultures of Research, RWTH, Aachen, Germany; ehsan.nabavi@anu.edu.au

More critical work

More critical work

Debunking the concept of digital twins of the Earth

The Honest Broker

Digital Twins of the Earth: Science or Pseudoscience?

A guest post by Andrea Saltelli

ROGER PIELKE JR.
SEP 09, 2024

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WIREs CLIMATE CHANGE

RMetS Royal Geographical Society

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Bring digital twins back to Earth

Andrea Saltelli, Gerd Gigerenzer, Mike Hulme, Konstantinos V. Katsikopoulos, Lieke A. Melsen, Glen P. Peters, Roger Pielke Jr, Simon Robertson, Andy Stirling, Massimo Tavoni, Arnald Puy

First published: 26 August 2024 | https://doi.org/10.1002/wcc.915 | VIEW METRICS

Edited by Matthias Heymann, Domain Editor and Maria Carmen Lemos, Editor-in-Chief

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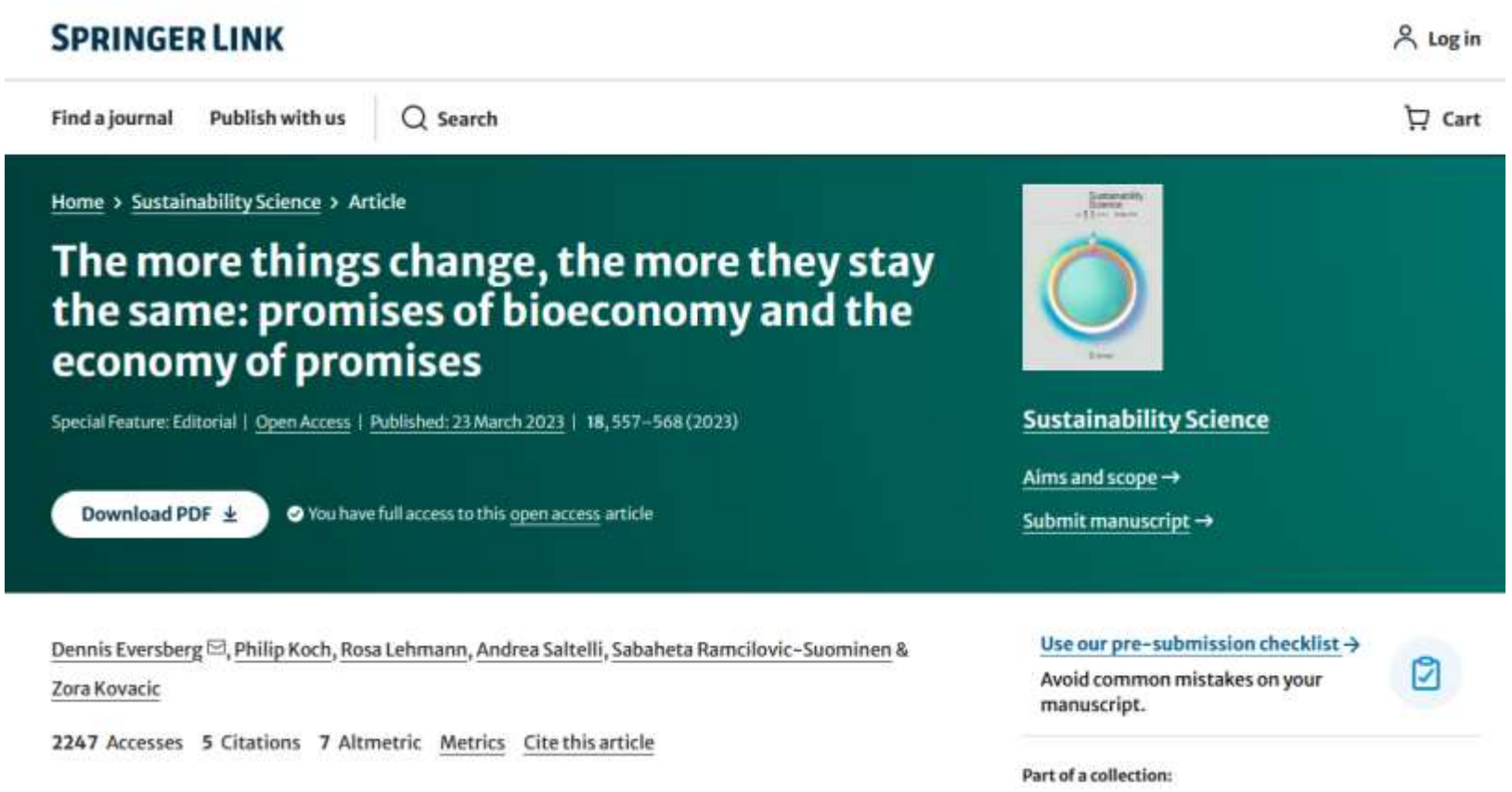

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